learn, MAB and BayesOpt are very similar problems—both deal with decision making under uncertainty for optimization—so the expectation is that solutions to the MAB problem will work well on BayesOpt too.

First, we will briefly talk about the MAB problem and how it is related to BayesOpt. This discussion will provide some background on the problem and put into perspective how BayesOpt is related to other problems in artificial intelligence. Then, we will learn about the two most popular policies in MAB and apply them to BayesOpt. The first is the *Upper Confidence Bound* policy, which uses the "optimism in the face of uncertainty" principle to reason about its decisions. The second policy is called *Thompson sampling*, which is a randomized solution that actively leverages the probabilistic nature of our predictive models. We then implement and run these policies on our running examples and analyze their performance.

By the end of this chapter, we will gain an understanding of the MAB problem and its relationship with BayesOpt. More importantly, we will add two more items to our portfolio of BayesOpt policies, exposing ourselves to more ways of exploring the search space in a black-box optimization problem.

5.1 Introduction to the multi-armed bandit problem

In this section, we learn about the multi-armed bandit (MAB) problem on a high level, including its problem statement and setting in the first subsection.

> **IMPORTANT** **In a multi-armed bandit problem, we need to choose an action to take at each step over a long horizon. Each action yields reward according to an unknown reward rate, and our goal is to maximize the total reward we receive at the end of the long horizon.**

We will also explore its connections to Bayesian optimization (BayesOpt) as well as other problems in artificial intelligence and machine learning. This will provide us with context of why MAB is relevant to our topic.

5.1.1 Finding the best slot machine at a casino

While "multi-armed bandit" may evoke a mysterious image in your mind, the term actually refers to a gambler’s problem of choosing which slot machines to play at a casino in order to obtain the largest amount of reward. Imagine you’re at a casino and there’s a slot machine with an *arm* that you can pull.

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